SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO4	**Assessment:**	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	**Total Marks:**	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	Krishna Tejesh Reddy .Y	**Reg. No.:**	192411200
Date:	26-8-2026	**Duration:**	60 minutes

Code of Conduct

I _ Krishna Tejesh Reddy .Y *(192411200) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

YKrishnatejeshReddy

Signature of the Student: __ ______________________

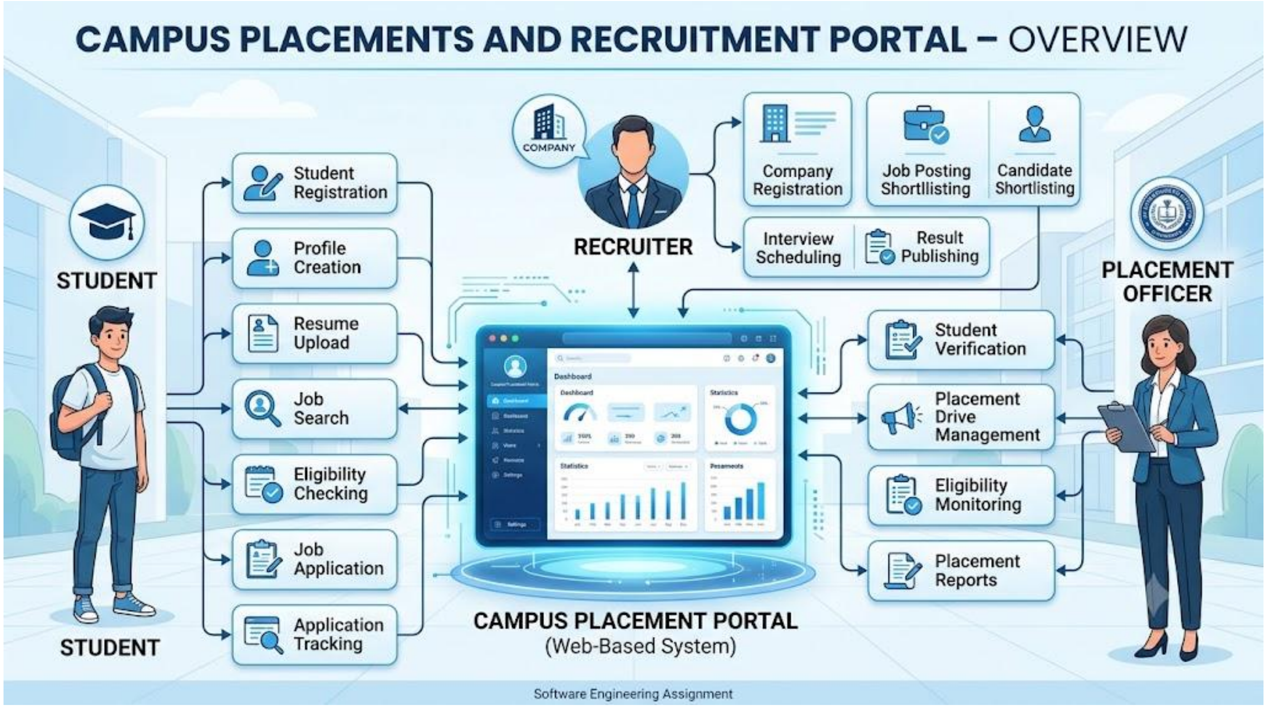
Annexure A – Test Case Design Assignment

1. Introduction

A **Campus Placements and Recruitment Portal** is a web-based software system designed to automate and manage the campus recruitment process. The portal connects students, placement officers, and recruiting companies through a centralized platform.

Students can register, create and update their profiles, upload resumes, view eligible job opportunities, apply for recruitment drives, and track their application status. Recruiters can register their organizations, create job openings, define eligibility criteria, shortlist candidates, schedule interviews, and publish selection results. Placement officers can manage student records, verify eligibility, coordinate recruitment drives, and monitor placement activities.

Testing this system is important because incorrect eligibility calculations, application failures, inaccurate student information, or unauthorized access can directly affect placement opportunities. Therefore, the portal must be tested for **functionality, reliability, usability, security, performance, and data integrity**.



2. Objectives of Testing

The main objectives of testing the Campus Placements and Recruitment Portal are:

1. To verify that students can successfully register and log in.
2. To verify that student profiles and resumes are stored correctly.
3. To verify that job opportunities are displayed correctly.
4. To verify that eligibility criteria are evaluated accurately.
5. To verify that eligible students can apply for recruitment drives.
6. To verify that ineligible students cannot apply when restrictions are applicable.
7. To verify that recruiters can create and manage job postings.
8. To verify that recruiters can shortlist eligible candidates.
9. To verify that interviews can be scheduled correctly.
10. To verify that placement results are published accurately.
11. To verify that unauthorized users cannot access restricted information.
12. To verify the reliability, usability, performance, and security of the portal.

3. System Modules

The Campus Placements and Recruitment Portal can be divided into the following major modules:

3.1 Student Module

- Student registration
- Student login
- Profile management
- Academic details
- Resume upload
- Job search
- Eligibility checking
- Job application
- Application tracking
- Interview information
- Selection status

3.2 Recruiter Module

- Recruiter/company registration
- Recruiter login
- Company profile
- Job posting
- Eligibility criteria definition
- Candidate viewing
- Candidate shortlisting
- Interview scheduling
- Selection result submission

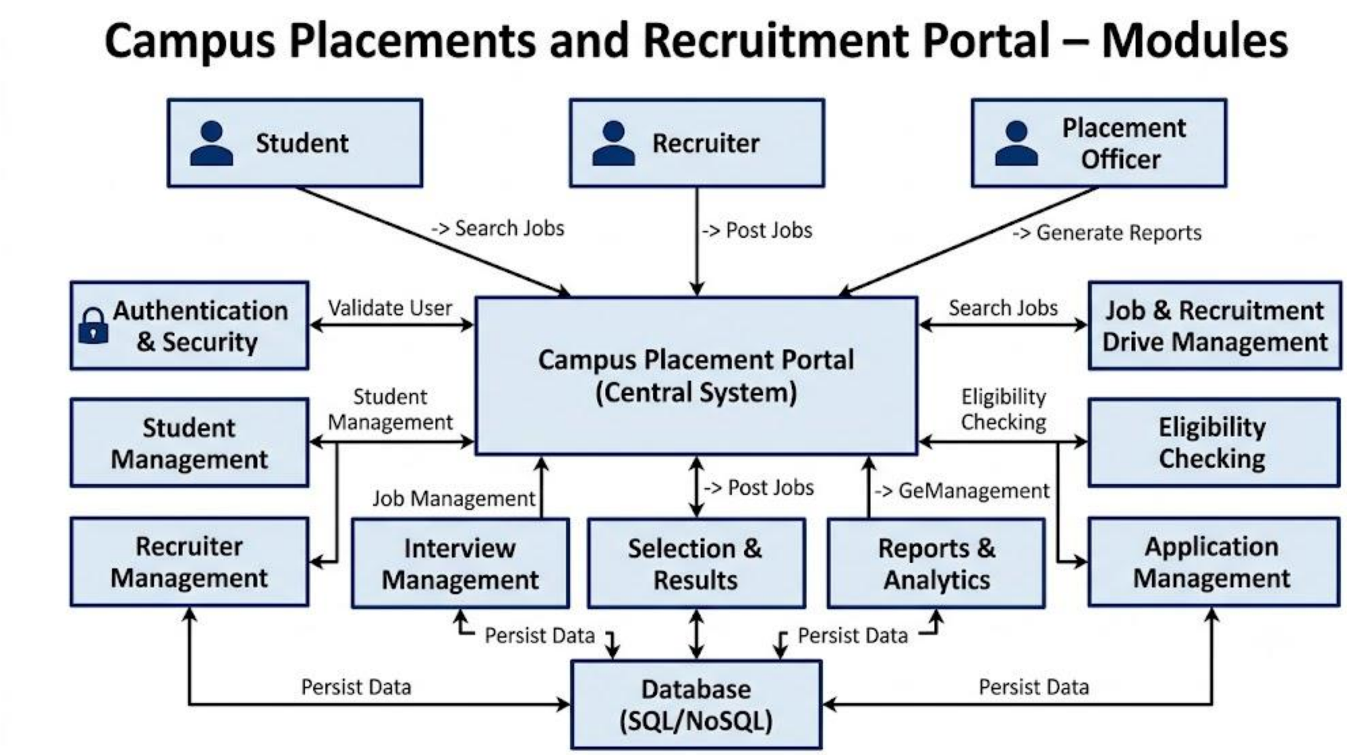
3.3 Placement Officer Module

- Student verification
- Recruiter verification
- Recruitment-drive management
- Eligibility monitoring

- Application monitoring
- Placement statistics
- Report generation

3.4 Authentication and Security Module

- Login authentication
- Password validation
- Role-based access
- Session management
- Unauthorized-access prevention



4. Functional Requirements

The following functional requirements need to be tested.

ID	Functional Requirement
FR01	The system shall allow students to register.
FR02	The system shall authenticate registered users.
FR03	The system shall allow students to create and update profiles.
FR04	The system shall allow students to upload resumes.
FR05	The system shall display available recruitment opportunities.
FR06	The system shall determine student eligibility based on defined criteria.
FR07	The system shall allow eligible students to apply for jobs.
FR08	The system shall prevent duplicate applications where applicable.
FR09	The system shall allow recruiters to create job postings.
FR10	The system shall allow recruiters to specify eligibility criteria.
FR11	The system shall allow recruiters to shortlist candidates.
FR12	The system shall allow interview schedules to be created.
FR13	The system shall allow selection results to be recorded.
FR14	The system shall allow students to view application status.
FR15	The system shall allow placement officers to monitor recruitment activities.
FR16	The system shall generate placement-related reports.

5. Non-Functional Requirements

The following non-functional requirements should also be tested.

ID	Requirement	Testing Focus
NFR01	Performance	Pages should load within an acceptable time.
NFR02	Security	Unauthorized users should not access restricted modules.
NFR03	Usability	Interface should be simple and understandable.
NFR04	Reliability	System should function consistently without unexpected failures.
NFR05	Availability	Portal should remain accessible during recruitment periods.

NFR06	Data Integrity	Student and recruitment data should remain accurate.
NFR07	Compatibility	Portal should work across supported browsers and devices.
NFR08	Scalability	System should support many students and recruiters simultaneously.
NFR09	Maintainability	System should be easy to update and maintain.
NFR10	Privacy	Student and recruiter information should be protected.

6. Test Scenarios

Test scenarios identify the major situations that need to be tested.

Scenario ID	**Test Scenario**
TS01	Verify student registration with valid details.
TS02	Verify registration with an existing email ID.
TS03	Verify login using valid credentials.
TS04	Verify login using invalid credentials.
TS05	Verify student profile creation.
TS06	Verify profile update.
TS07	Verify resume upload with valid file.
TS08	Verify resume upload with invalid file type.
TS09	Verify job posting by recruiter.
TS10	Verify eligibility calculation.
TS11	Verify application by eligible student.
TS12	Verify application by ineligible student.
TS13	Verify duplicate application prevention.
TS14	Verify recruiter candidate shortlisting.
TS15	Verify interview scheduling.

TS16	Verify student application-status tracking.
TS17	Verify selection result publishing.
TS18	Verify unauthorized page access.
TS19	Verify report generation.
TS20	Verify system behavior under high user load.

7. Test Case Design

The assignment specifically requires test cases to contain Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status**.**

For the Actual Result and Pass/Fail columns, enter them during actual execution. For a design assignment, you can initially use "To be executed" and "Not Executed"**.**

7.1 Student Registration Test Cases

TC ID	**Test Scenario**	**Test Steps**	**Test Data**	**Expected Result**	**Actual Result**	**Status**
TC01	Valid student registration	Open registration → enter details → submit	Valid name, register number, email, password	Account should be created successfully	To be executed	Not Executed
TC02	Existing email	Enter already registered email → submit	Existing email	System should display duplicate-email error	To be executed	Not Executed

TC03	Invalid email	Enter invalid email → submit	student@	Validation message should appear	To be executed	Not Executed
TC04	Empty mandatory fields	Leave required fields empty → submit	Blank fields	Required-field messages should appear	To be executed	Not Executed
TC05	Invalid register number	Enter invalid register number	ABC123	System should reject invalid format	To be executed	Not Executed

8. Login Test Cases

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC06	Valid login	Enter valid username/password → Login	Valid credentials	User should reach dashboard	To be executed	Not Executed
TC07	Wrong password	Enter valid email and wrong password	Wrong password	Error message should appear	To be executed	Not Executed
TC08	Unregistered user	Enter unregistered email	Unknown email	Login should fail	To be executed	Not Executed
TC09	Empty credentials	Leave fields empty → Login	Blank username/password	Validation messages should appear	To be executed	Not Executed
TC10	Password masking	Enter password	Campus@123	Password should be	To be executed	Not Executed

				hidden/masked		

9. Student Profile and Resume Test Cases

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC11	Create profile	Enter academic and personal details → Save	Valid details	Profile should be saved	To be executed	Not Executed
TC12	Update profile	Modify profile → Save	Updated phone number	Updated information should be stored	To be executed	Not Executed
TC13	Valid resume upload	Select supported resume → Upload	PDF resume	Resume should upload successfully	To be executed	Not Executed
TC14	Invalid resume format	Select unsupported file → Upload	.exe file	Upload should be rejected	To be executed	Not Executed
TC15	Missing resume	Apply without required resume	No resume	System should show appropriate validation	To be executed	Not Executed

10. Eligibility Testing

Assume a recruitment drive has the following criteria:

- Minimum CGPA: **7.0**
- Maximum active backlogs: **0**
- Required graduation branch: **CSE/IT**

TC ID	Test Scenario	Test Data	Expected Result
TC16	Student exactly meets CGPA requirement	CGPA = 7.0, Backlogs = 0	Student should be eligible
TC17	Student below CGPA requirement	CGPA = 6.9, Backlogs = 0	Student should be marked ineligible
TC18	Student exceeds CGPA requirement	CGPA = 8.5, Backlogs = 0	Student should be eligible
TC19	Student has one backlog	CGPA = 8.0, Backlogs = 1	Student should be marked ineligible
TC20	Wrong branch	CGPA = 8.0, Branch = Mechanical	Student should be marked ineligible

11. Job Application Test Cases

TC ID	Test Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC21	Eligible student applies	Open eligible job → Apply	Eligible student	Application should be submitted	To be executed	Not Executed
TC22	Ineligible student applies	Open restricted job → Apply	CGPA below requirement	Application should be blocked	To be executed	Not Executed
TC23	Duplicate application	Apply twice for same job	Same student/job	Second application should be prevented	To be executed	Not Executed
TC24	Application deadline passed	Apply after deadline	Expired job	Application should not be accepted	To be executed	Not Executed

TC25	Missing profile information	Apply with incomplete profile	Missing academic data	System should request profile completion	To be executed	Not Executed

12. Recruiter Test Cases

TC ID	Test Scenario	Test Steps	Test Data	Expected Result
TC26	Create job posting	Login → Create Job → Enter details → Submit	Valid job details	Job should be published
TC27	Missing job information	Leave required field blank	Missing salary/criteria	Validation should appear
TC28	Set eligibility criteria	Enter CGPA/branch/backlog criteria	CGPA 7.0, CSE	Criteria should be saved
TC29	Shortlist candidate	Select candidate → Shortlist	Eligible candidate	Candidate status should change to shortlisted
TC30	Schedule interview	Select candidate → Schedule	Date/time/venue	Interview should be scheduled

13. Application Status and Selection Test Cases

TC ID	Test Scenario	Test Steps	Expected Result
TC31	View submitted application	Student opens My Applications	Submitted application should be displayed
TC32	View shortlisted status	Recruiter shortlists student	Student should see “Shortlisted”
TC33	View rejected status	Recruiter rejects student	Student should see “Rejected”
TC34	Publish selection result	Recruiter selects candidate	Result should be recorded
TC35	Student views selection	Open application status	Student should see selection status

14. Security Test Cases

TC ID	Test Scenario	Test Steps	Expected Result
TC36	Unauthorized dashboard access	Student attempts recruiter URL	Access should be denied
TC37	Unauthorized admin access	Student attempts officer page	Access should be denied
TC38	Session logout	Logout → press Back	Protected pages should not be accessible
TC39	Invalid password	Enter incorrect password repeatedly	Login should remain protected
TC40	Password visibility	Enter password	Password should remain masked

15. Test Design Techniques

The assignment expects appropriate techniques such as **Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing**.

15.1 Equivalence Partitioning

Equivalence Partitioning divides input data into valid and invalid groups.

Example: CGPA

Assume minimum eligible CGPA = **7.0**.

Partition	Input	Expected Result
Invalid	0–6.9	Ineligible
Valid	7.0–10.0	Eligible
Invalid	Above 10.0	Invalid input

This reduces the number of test cases while still testing representative values.

15.2 Boundary Value Analysis

Boundary Value Analysis tests values at and around the limits.

For minimum CGPA = **7.0**, test:

Test Value	Expected Result
6.9	Ineligible
6.99	Ineligible
7.0	Eligible
7.01	Eligible

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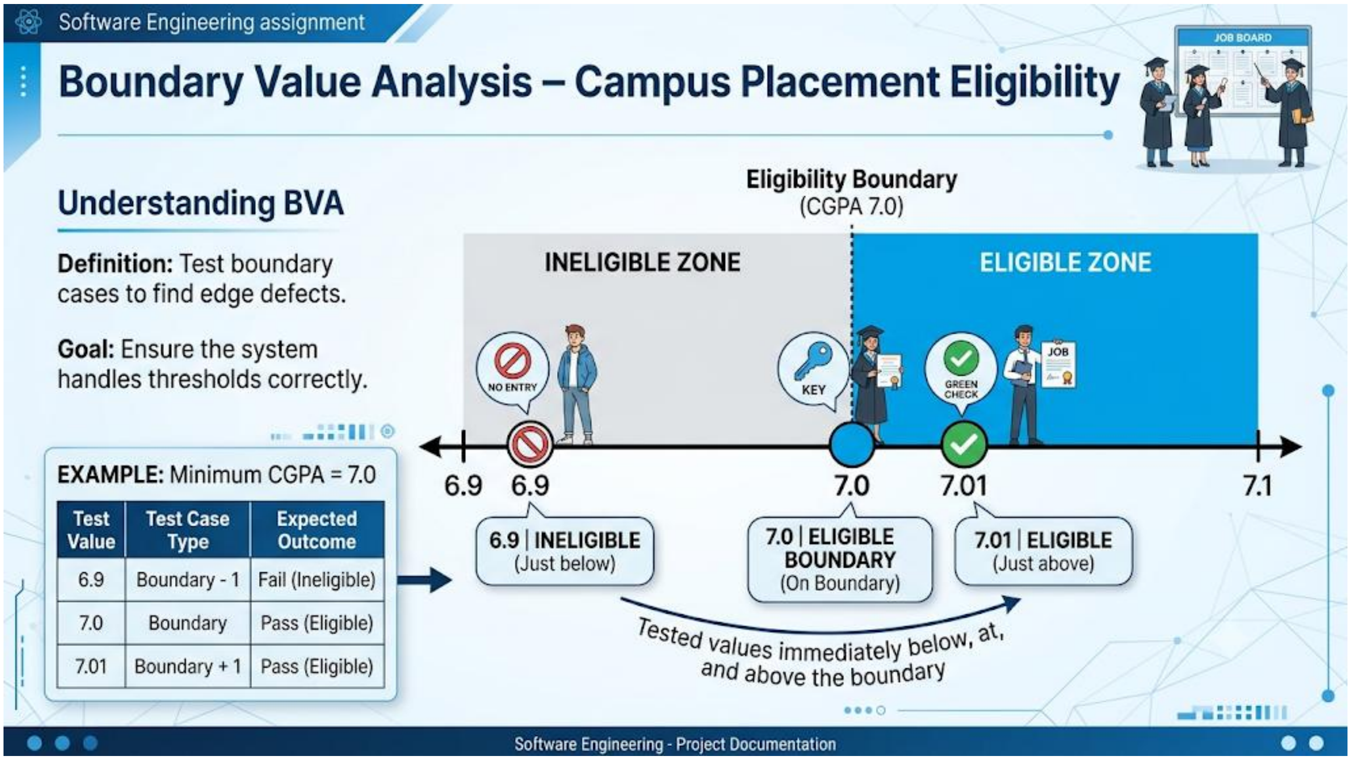


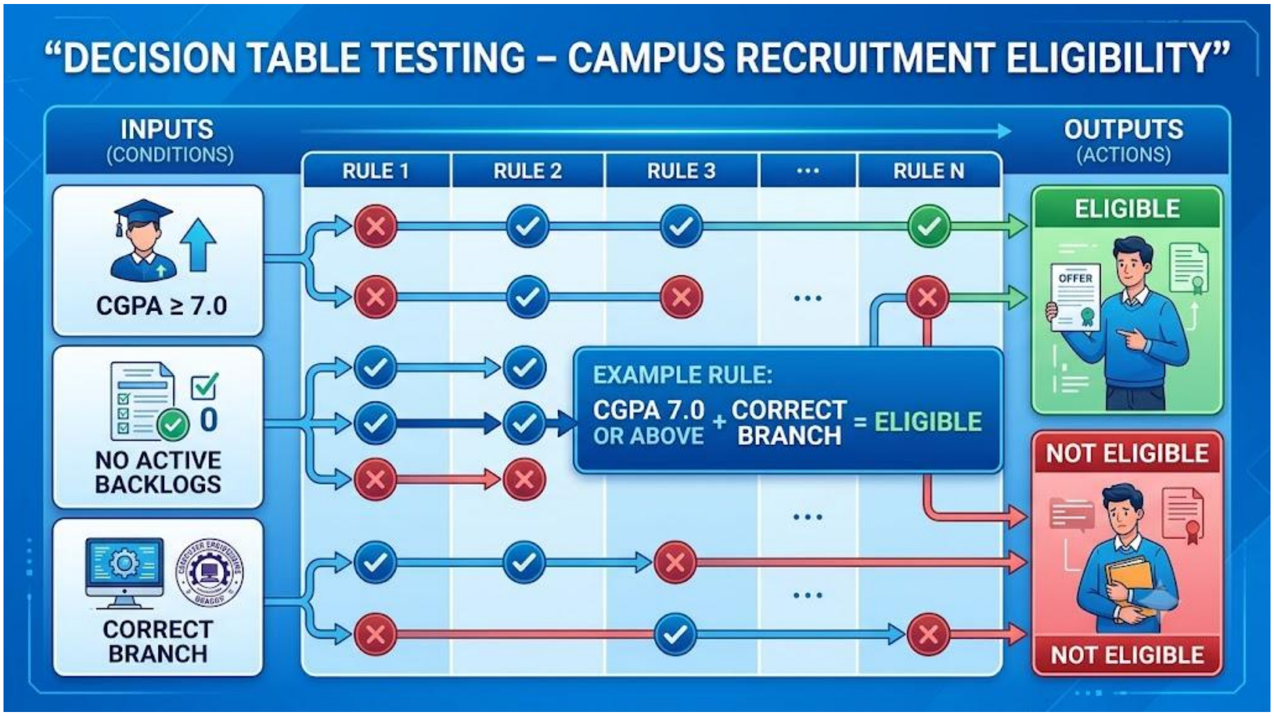
16. Decision Table Testing

Decision Table Testing can be used for determining whether a student is eligible for a recruitment drive.

Conditions

- CGPA ≥ 7.0
- No active backlogs
- Eligible branch

Rule	CGPA ≥ 7.0	No Backlogs	Correct Branch	Result
R1	Yes	Yes	Yes	Eligible
R2	No	Yes	Yes	Not Eligible
R3	Yes	No	Yes	Not Eligible
R4	Yes	Yes	No	Not Eligible
R5	No	No	Yes	Not Eligible
R6	No	Yes	No	Not Eligible
R7	Yes	No	No	Not Eligible
R8	No	No	No	Not Eligible



17. State Transition Testing

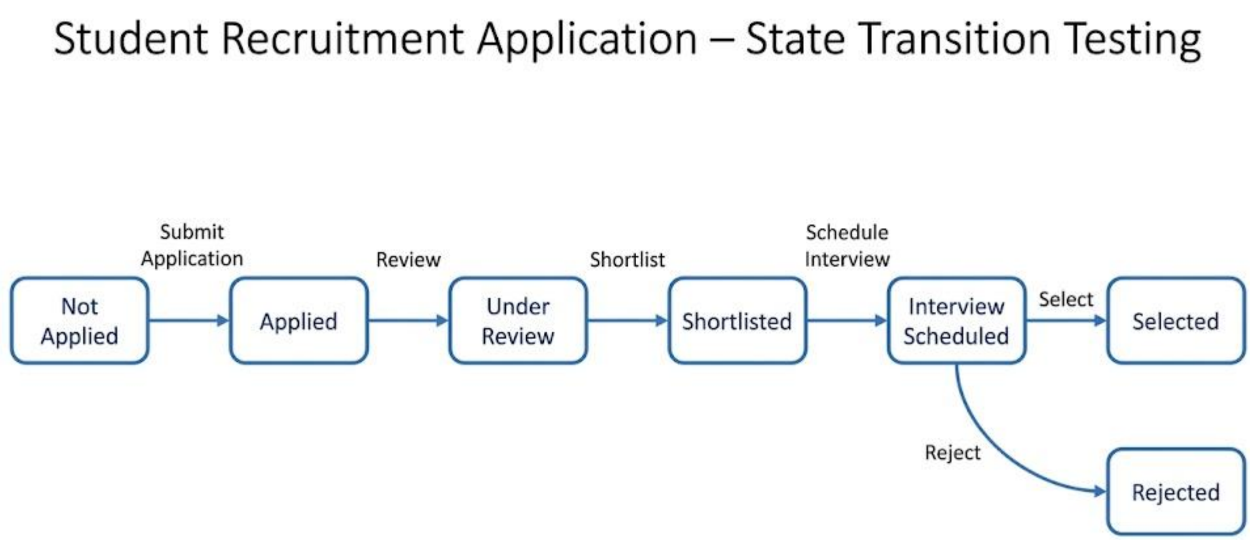
State Transition Testing can be applied to the **student recruitment application lifecycle**.

A student's application may move through these states:

Not Applied → Applied → Under Review → Shortlisted → Interview Scheduled → Selected/Rejected

Example:

Current State	**Event**	**Next State**
Not Applied	Submit application	Applied
Applied	Recruiter reviews	Under Review
Under Review	Candidate selected for next stage	Shortlisted
Shortlisted	Interview scheduled	Interview Scheduled
Interview Scheduled	Candidate selected	Selected
Interview Scheduled	Candidate rejected	Rejected



18. Non-Functional Testing

18.1 Performance Testing

The portal should be tested under different numbers of concurrent users.

Example:

- 10 users
- 100 users
- 500 users
- 1000 users

The objective is to verify that important pages such as login, job listings, and application submission remain responsive.

18.2 Security Testing

Security testing should verify:

- Authentication
- Authorization
- Password protection
- Session management
- Role-based access
- Protection against unauthorized access
- Protection of student information

18.3 Usability Testing

The system should be easy for students, recruiters, and placement officers to understand and use.

Test areas include:

- Navigation
- Form clarity

- Error messages
- Button visibility
- Mobile responsiveness
- Consistent interface

18.4 Compatibility Testing

The portal should be tested on supported:

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Desktop screens
- Laptop screens
- Mobile devices

19. Exceptional and Negative Test Cases

Testing should not be limited to normal conditions. The following exceptional situations should also be considered:

1. Internet connection is interrupted during application submission.
2. Server becomes unavailable while uploading a resume.
3. User session expires during form submission.
4. Student attempts to submit an application after the deadline.
5. Recruiter enters incomplete job information.
6. Student uploads a corrupted resume.
7. Multiple users attempt to access the system simultaneously.
8. Database connection becomes temporarily unavailable.
9. User refreshes the page during an application transaction.
10. Unauthorized user attempts to access restricted information.

These tests help verify the **reliability and robustness** of the portal.

20. Requirement Traceability Matrix

Traceability provides mapping between requirements, scenarios, and test cases. This directly supports the assignment's requirement for strong requirement-to-test coverage.

Requirement	Test Scenario	Test Cases
FR01 Student Registration	TS01	TC01–TC05
FR02 Authentication	TS03, TS04	TC06–TC10
FR03 Profile Management	TS05, TS06	TC11–TC12
FR04 Resume Upload	TS07, TS08	TC13–TC15
FR05 Job Display	TS09	TC21
FR06 Eligibility	TS10	TC16–TC20
FR07 Job Application	TS11–TS13	TC21–TC25
FR09 Job Posting	TS09	TC26–TC28
FR11 Shortlisting	TS14	TC29
FR12 Interview	TS15	TC30
FR14 Status Tracking	TS16	TC31–TC35
Security	TS18	TC36–TC40

21. Test Coverage

The designed test cases provide coverage across the major functional areas of the Campus Placements and Recruitment Portal.

Testing Area	Covered Test Cases
Registration	TC01–TC05
Authentication	TC06–TC10
Profile Management	TC11–TC12
Resume Management	TC13–TC15
Eligibility	TC16–TC20
Applications	TC21–TC25
Recruiter Functions	TC26–TC30

Application Status	TC31–TC35
Security	TC36–TC40

22. Summary of Test Design Techniques

Technique	**Application in Portal**
Equivalence Partitioning	CGPA, backlog count, registration inputs
Boundary Value Analysis	Minimum CGPA, maximum allowed values, application deadline
Decision Table Testing	Student eligibility determination
State Transition Testing	Application status lifecycle
Negative Testing	Invalid login, invalid resume, incomplete forms
Security Testing	Unauthorized access and authentication
Performance Testing	Multiple concurrent users
Compatibility Testing	Different browsers/devices

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23. Conclusion

The Campus Placements and Recruitment Portal requires systematic testing because it manages important student, recruiter, and placement information. The proposed test cases cover the major functional and non-functional requirements of the system, including registration, authentication, profile management, resume upload, eligibility checking, job applications, recruiter activities, interviews, selection results, and security.

The use of **Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing** improves the effectiveness of the testing process by covering normal, invalid, boundary, and state-based conditions.

A comprehensive testing approach helps identify defects early, improves system reliability and usability, protects sensitive information, and ensures that students and recruiters receive accurate and dependable placement services.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
Test Case Design & Completeness(25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	**Mark**
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25